

MAC 2025: The 2nd Micro-Action Analysis Grand Challenge

Kun Li
ReLER, CCAI, Zhejiang University
Hangzhou, China
kunli.hfut@gmail.com

Dan Guo*
Hefei University of Technology
Institute of Artificial Intelligence,
Hefei Comprehensive National
Science Center
Hefei, China
guodan@hfut.edu.cn

Xiaobai Li
Zhejiang University
University of Oulu
Hangzhou, China
xiaobai.li@zju.edu.cn

Haoyu Chen
CMVS, University of Oulu
Oulu, Finland
chen.haoyu@oulu.fi

Pengyu Liu
Hefei University of Technology
Hefei, China
lpynow@gmail.com

Fei Wang
Hefei University of Technology
Hefei, China
jiafei127@gmail.com

Jingjing Hu
Hefei University of Technology
Hefei, China
xianhjj623@gmail.com

Guoying Zhao
CMVS, University of Oulu
Oulu, Finland
guoying.zhao@oulu.fi

Meng Wang
Hefei University of Technology
Hefei, China
eric.mengwang@gmail.com

Abstract

Micro-Actions (MAs) are a crucial form of non-verbal communication in social interactions, with promising applications in human emotion analysis. Although the topic has attracted considerable research interest, progress has been hindered by the lack of publicly available benchmark datasets. To address this gap, the Micro-Action Analysis Grand Challenge (MAC) is organized annually. This paper presents an overview of the 2nd Micro-Action Analysis Grand Challenge, held in conjunction with ACM Multimedia 2025. We provide a comprehensive summary of the challenge, including its dataset, evaluation protocol, results, and discussion. The top-ranked solutions are highlighted to offer valuable insights for researchers, and potential future directions are outlined to guide ongoing developments in this area. The goal of this grand challenge is to foster innovative research in micro-action analysis and advance research in the human-centric action understanding community.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Activity recognition and understanding**.

Keywords

Micro-Action Recognition, Multi-label Micro-Action Detection

*Corresponding author

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MM '25, Dublin, Ireland

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ACM ISBN 979-8-4007-2035-2/2025/10
<https://doi.org/10.1145/3746027.3762086>

ACM Reference Format:

Kun Li, Dan Guo, Xiaobai Li, Haoyu Chen, Pengyu Liu, Fei Wang, Jingjing Hu, Guoying Zhao, and Meng Wang. 2025. MAC 2025: The 2nd Micro-Action Analysis Grand Challenge. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3746027.3762086>

1 Introduction

As an important non-verbal communicative fashion, human body actions [6] are capable of conveying emotional information during social communication. In previous works, efforts have been made mainly on facial expressions [37], speech [38], or expressive body gestures to interpret classical expressive emotions. Differently, we focus on a specific group of body actions, called Micro-Actions (MAs) [4, 6, 10, 22], used in the psychology research field to interpret inner human feelings. Recently, there has been a lot of interest in empowering intelligent machines with the capabilities for understanding human behaviors [2, 6, 10, 22]. Micro-action behavior analysis [5, 9, 10, 14, 15, 20] has become an important research topic due to its potential applications in human-to-human communication and human emotion state analysis.

Although micro-action analysis plays a crucial role in fine-grained human behavior understanding, it remains an underexplored area in the broader vision community [24, 28, 29, 33, 39]. Compared to conventional action recognition tasks [16, 19, 30], this field is still in its early stages of development, due to the lack of accessible entry points for new researchers. To bridge this gap, we have launched the 1st Micro-Action Analysis Challenge (MAC)¹ [7, 10, 18, 27, 36] in conjunction with ACM MM 2024, with the goal of providing high-quality benchmark data and a fair, unified platform for performance comparison. The MAC Challenge is designed as an annual series with a consistent yet evolving theme to drive continuous progress in the field. Building upon this foundation, the 2nd edition of MAC²,

¹1st MAC: <https://sites.google.com/view/micro-action2024>.

²2nd MAC: <https://sites.google.com/view/micro-action>.

held at ACM MM 2025, marks a significant step forward. Through this initiative, we seek to promote broader participation and inspire the research community to further investigate and advance the core challenges of micro-action analysis.

2 Challenge Overview

2.1 Challenge Tracks

This year, we also set two parallel tracks for this grand challenge:

□ **Track 1: Micro-Action Recognition (MAR).** MAR [8, 10, 11, 13] aims to recognize and distinguish subtle body actions that typically occur in a brief instant. MAR shares similarities with conventional action recognition [3, 32], as it involves processing video instances and demands both accuracy and efficiency. However, MAR presents unique challenges due to its subtle motion amplitudes and high inter-class similarity. In this track, teams are allowed to use raw video data to develop models for recognizing both body-level and action-level categories.

□ **Track 2: Multi-label Micro-Action Detection (MMAD).** Considering the co-occurrence of human micro-actions, *i.e.*, the same micro-action may be repeated in time and different micro-actions may occur at the same time, we propose the Multi-label Micro-Action Detection (MMAD) task [15] to facilitate a deeper understanding of human bodily behavior. MMAD refers to the task of identifying and localizing all micro-actions in a given densely annotated video, determining their corresponding start and end times, as well as their micro-action categories.

2.2 Data

The data used for the 2nd Micro-Action Analysis Grand Challenge comes from the MA-52 dataset [10] and MMA-52 dataset [15].

□ **Track 1: Micro-Action Recognition.** MA-52 [10] is a large-scale, whole-body micro-action dataset collected by professional interviewers to capture unconscious human micro-behaviors. It comprises 22,422 samples from 205 participants, with annotations organized into two hierarchical levels: 7 body-level categories and 52 fine-grained action-level categories. It is worth noting that all training and validation data from the MA-52 dataset are utilized in Track 1. For evaluation, 1,138 samples were selected from the MA-52 test set to serve as the challenge’s test data. Detailed statistics are provided in Table 2. Final rankings are determined based on performance on this test set.

□ **Track 2: Multi-label Micro-Action Detection.** In this track, the training, validation, and test data are the same as the MMA-52 dataset [15]. Table 2 presents the statistics of the MMA-52 dataset. Note that this challenge only evaluates performance at the action-level for micro-action detection. The final rankings are based on detection performance on the test set.

2.3 Challenge Protocol

The 2nd MAC challenge is launched on the CodaLab platform [23], consisting of two stages as follows.

□ **Development Stage (26.03.2025 - 01.07.2025)** The 2nd MAC Challenge was officially launched on March 26th, 2025, with the tasks for both tracks announced simultaneously. The training and validation datasets were released at the same time. Participants were allowed to register either individually or as a team, with the

Table 1: Data statistics for the data used in Track 1. “Avg. Video” denotes the average length of videos.

Split	Videos	Duration	Avg. Video
Training	11,250	6.19h	1.98s
Validation	5,586	3.05h	1.96s
Test	1,138	0.71h	2.23s
All	17,974	9.95h	2.06s

Table 2: Data statistics for the data used in Track 2. “Duration” refers to the length of all videos, and “Avg. Insta.” represents the average length of action instances.

Split	Videos	Instances	Duration	Avg. Video	Avg. Insta.	#Subj.
Training	4,534	13,698	12.91h	10.25s	4.10s	140
Validation	1,475	4,735	4.34h	10.60s	4.10s	37
Test	519	1,349	1.42h	9.86s	3.79s	26
All	6,528	19,782	18.67h	10.30s	4.07s	203

registration deadline set for July 1st, 2025. All registered teams were required to submit their team information, and to ensure fairness, each team was assigned a single authorized account for result submission. To access the challenge data, participating teams were also required to sign a license agreement file in advance.

□ **Testing Stage (02.07.2025 - 09.07.2025)** The testing data for both tracks was released on July 1st, 2025. All participating teams are allowed to submit their testing results on CodaLab between July 2nd and July 9th, 2025. To ensure fairness in the competition, each team must submit results using a pre-authorized account. Each account is allowed a maximum of 35 submissions in total, with a daily limit of 5 submissions. The leader board ranking will be automatically updated based on the submitted results.

2.4 Evaluation Metrics

□ **Track 1: Micro-Action Recognition.** Given the diversity of micro-actions across both body-level and action-level categories, we use the F1 score to evaluate the performance of micro-action recognition. $F1_{macro}$ calculates the unweighted mean for each action category, independent of the sample size for the specific category. $F1_{micro}$ treats all samples equally and is less influenced by any category with a predominant number of micro-actions. We use $F1_{mean}$ as the evaluation metric:

$$F1_{mean} = \frac{F1_{macro}^{body} + F1_{micro}^{body} + F1_{macro}^{action} + F1_{micro}^{action}}{4}, \quad (1)$$

where “body” and “action” denote the label granularity of body-level and action-level categories, respectively. The final results are ranked by the value of $F1_{mean}$.

□ **Track 2: Micro-Action Recognition.** We adopt the Detection-mAP [15, 21] to measure the completeness of predicted micro-action instances. It calculates the mean Average Precision (mAP) of action predictions at different tIoU thresholds, providing an instance-wise evaluation. We report the average mAP results for tIoU thresholds ranging from 0.1 to 0.9 in increments of 0.1 and the average mAP. The higher the mAP, the better the micro-action detection. The final results are ranked by the value of average mAP.

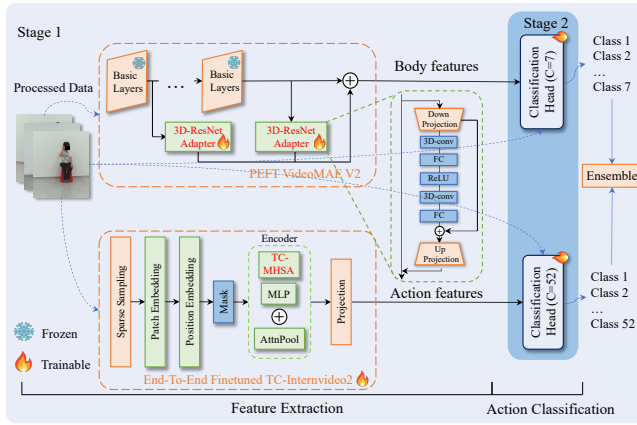


Figure 1: Overall framework for the team USTC-IAT-United in Track 1.

3 Proposed Approaches

A total of 53 teams registered for the challenge, with 46 teams participating in Track 1 and 35 teams in Track 2. We contacted the top three teams in each track to request brief descriptions of their methods for inclusion in this summary paper. With their permission, these descriptions are presented below.

3.1 Track 1: Micro-Action Recognition

① Team “USTC-IAT-United” participated in Track 1, and they came from the University of Science and Technology of China [34]. As illustrated in Figure 1, the team’s approach adopted a two-stage framework comprising feature extraction and action classification, specifically tailored for the micro-action recognition task.

For body-action feature extraction, the team designed a 3D-ResNet Adapter module to enhance the parameter-efficient fine-tuning (PEFT) capabilities of VideoMAE v2 [31]. The adapter is composed of two 3D convolutional layers, two fully connected layers, and a nonlinear activation operated as the sole trainable component. In terms of micro-action feature modeling, the team leveraged end-to-end fine-tuning on InternVideo2 and introduced a novel Temporal Contextualization (TC) mechanism. By replacing the standard Multi-Head Self-Attention (MHSA) module with a TC-MHSA variant, the system strengthened its capacity to integrate multi-frame spatiotemporal cues and semantically rich contextual information. For the downstream action classification, a TimeS-former head was employed. To enhance robustness against class imbalance and training instability, the loss function combined a weighted cross-entropy loss with a locality-aware auxiliary loss. Finally, the team adopted a weighted ensemble strategy that fused the outputs of the coarse-grained and fine-grained branches, improving the overall recognition accuracy across both body-level and action-level prediction tasks.

② Team “HIT” participated in Track 1, and they came from Harbin Institute of Technology (Weihai) and the University of Science and Technology of China [26]. This team proposed an end-to-end solution for micro-action recognition, effectively addressing

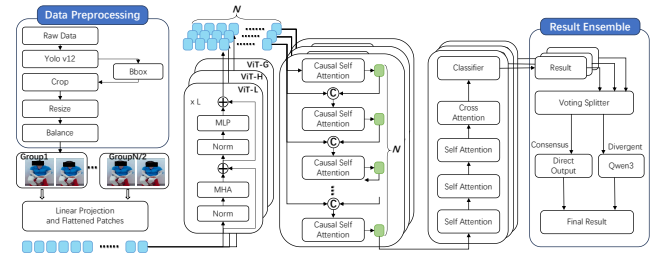


Figure 2: Overall framework for the team HIT in Track 1.

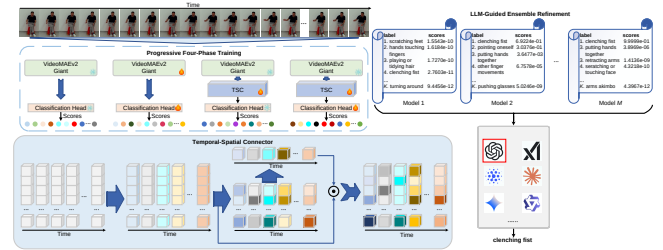


Figure 3: Overall framework for the team Wind of Innovation in Track 1.

spatial noise and long-tailed class distribution. Their pipeline included four key stages: pre-processing, feature learning, ensemble inference, and post-processing.

In the pre-processing stage, YOLOv12 [25] was used to crop and resize the person-centered region, enhancing action details while suppressing background noise. A dynamic oversampling strategy balanced the class distribution, followed by patch-based encoding for model input. Feature learning leveraged a frozen V-JEPA 2 [1] encoder and parallel Transformer branches (e.g., ViT-G/H/L), with a lightweight probe module that replaced standard self-attention with cross-attention using learnable query tokens for efficient aggregation. During inference, multiple base models (each with 20 classifiers) formed an ensemble. Final predictions were obtained through weighted voting based on validation performance, improving robustness. For post-processing, the team introduced prediction-space compression: high-consensus samples were directly accepted, while ambiguous ones were refined via top-5 prediction aggregation and re-ranked by a large language model (Qwen3 [35]).

③ Team “Wind of Innovation” participated in Track 1 and comprised members from Huazhong University of Science and Technology, the University of Science and Technology of China, the City College of New York, the City University of New York, and Chongqing Zhihuo Shiwu Technology Co., Ltd [17]. The team’s method introduced a large-scale, fine-grained micro-action recognition framework tailored for the MA-52 dataset, effectively tackling the challenges of subtle motion perception and short temporal durations characteristic of micro-actions. Their approach featured three key innovations:

□ **Large-Scale Transformer Backbone:** To enhance spatial-temporal representation capacity, they scaled up to the Giant version of VideoMAE-v2 [31], pre-trained on Kinetics-710.

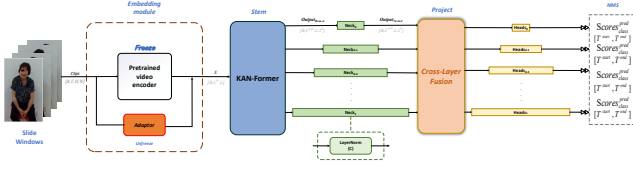


Figure 4: Overall framework for the team USTC-IAT-United of Innovation in Track 2.

□ **Temporal-Spatial Connector (TSC):** To address these challenges, they proposed a lightweight Temporal-Spatial Connector module for motion-sensitive feature recalibration. TSC effectively guided transformers to allocate representational capacity toward critical temporal segments and spatial regions, alleviating redundancy and improving sensitivity to fine-grained motion patterns.

□ **Progressive Four-Phase Training Strategy:** To ensure stable and efficient optimization, they adopted a progressive four-stage training paradigm: **1:** Linear probing, where the backbone remained frozen and only the classification head was trained for initial adaptation. **2:** Full fine-tuning of both the backbone and classification head to enable end-to-end representation learning. **3:** The Temporal-Spatial Connector was introduced, and only its parameters were optimized while the backbone and head remained frozen, preserving previously learned representations. **4:** The TSC and classification head were jointly refined, while the backbone remained fixed, facilitating alignment between motion-sensitive features and classification decision boundaries.

□ **LLM-Guided Ensemble Refinement:** To further enhance decision robustness, they proposed a large language model (LLM)-guided ensemble refinement strategy. Predictions from multiple diverse models were aggregated as structured textual prompts, which were processed by an LLM (e.g., GPT-4o [12]).

3.2 Track 2: Multi-label Micro-Action Detection

① Team “USTC-IAT-United” participated in Track 2, and they came from the University of Science and Technology of China. The team’s method addressed the core challenge in Temporal Action Detection (TAD) of capturing fine-grained spatiotemporal dependencies induced by micro-actions—motions characterized by low intensity and extremely short duration. To this end, the approach introduced an innovative cross-layer fusion framework built upon a feature pyramid architecture. At its core, the framework integrated two key components that significantly enhanced both perceptual and localization capabilities.

First, the method incorporated a novel module, KANFormer, as the fundamental unit for constructing the feature pyramid. KANFormer was specifically designed to model local, fine-grained spatiotemporal patterns, enabling more effective representation of the subtle and transient cues typical of micro-actions—areas where conventional architectures often fell short. **Second**, the approach proposed cross-layer fusion pathways, a standout innovation of the framework. These pathways enabled complementary feature integration across different pyramid levels: high-level features provided rich semantic context but with lower resolution and fuzzier boundaries, while low-level features offered precise spatiotemporal

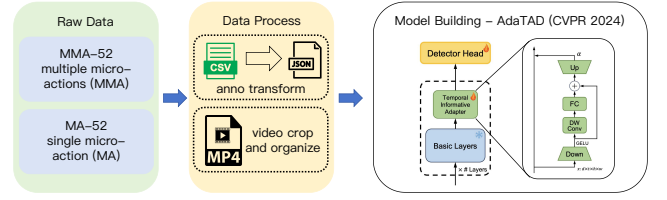


Figure 5: Overall framework for the team NJU-MCG in Track 2.

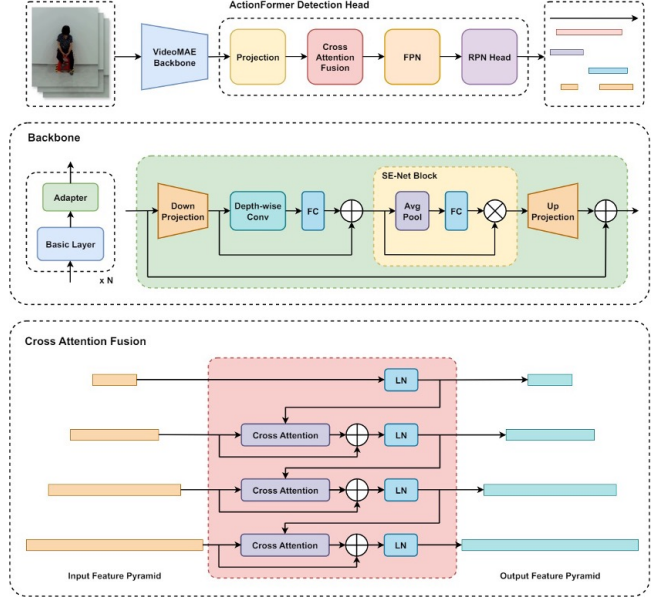


Figure 6: Overall framework for the team VIM in Track 2.

detail but weaker semantic abstraction. The fusion mechanism was designed to go beyond naive concatenation, facilitating mutual reinforcement between semantic abstraction and boundary precision. **As a result**, high-level features guided accurate recognition of complex actions and scenes, while low-level features sharpened localization accuracy, particularly at action boundaries. Through this synergistic design—combining the localized modeling strengths of KANFormer with semantically-aware cross-layer fusion—the method simultaneously advanced semantic recognition and precise boundary detection in micro-action scenarios.

② Team “NJU-MCG” participated in Track 1, and they came from Nanjing University. The team’s method was built upon AdaTAD [21]. To adapt AdaTAD effectively for this task, the team designed a targeted pipeline comprising three key components:

□ **Data preprocessing:** The team restructured the raw dataset, consisting of untrimmed videos and CSV-based annotations, into frame-aligned, segment-level JSONs tailored for AdaTAD.

□ **Task-Oriented Data Augmentation:** Given the limited size of the multi-action dataset (MMAD), the team introduced an augmentation strategy leveraging MA-52 from Track 1. Each annotated segment in MA-52 was repurposed as an individual micro-action

Table 3: The results of Track 1: Micro-Action Recognition. The results are provided by the Codalab competition page¹.

Rank	Team	Organization	Body Top-1	Action		Body		Action		All F1 _{mean}
				Top-1	Top-5	F1 _{macro}	F1 _{micro}	F1 _{macro}	F1 _{micro}	
1st	USTC-IAT-United	USTC	86.38	72.23	91.30	84.06	86.38	68.31	72.23	77.75
2nd	HIT	HIT & USTC	87.13	74.85	96.26	81.47	87.13	64.46	74.85	76.98
3rd	Wind of Innovation	HUST & USTC & CUNY	86.38	71.79	91.65	83.49	86.38	64.47	71.79	76.54

Table 4: The results of Track 2: Multi-label Micro-Action Detection. The results are provided by the Codalab competition page².

Rank	Team	Organization	mAP@0.2	mAP@0.5	mAP@0.7	avg. mAP
1st	USTC-IAT-United	USTC	40.62	32.06	16.24	27.04
2nd	NJU-MCG	NJU	38.24	26.25	14.55	25.15
3rd	VIM	NJUST	39.12	25.77	14.62	24.87

instance and integrated with MMAD to form an expanded and more diverse training set. This augmentation increased the dataset size approximately fivefold and enhanced coverage of rare or subtle motion patterns essential for micro-gesture recognition.

□ **Model Adaptation:** While AdaTAD provided a strong backbone, the team introduced several modifications to better align it with micro-action characteristics. They incorporated temporal adapters to focus on localized motion patterns and adjusted the temporal feature aggregation by tuning sampling strides and attention window sizes. These changes enabled the model to capture both short-range micro-movements and slightly longer contextual dependencies. Furthermore, they revised the positive label assignment strategy to accommodate overlapping action segments and employed a soft classification loss to improve tolerance to temporal imprecision—both of which contributed to greater robustness in densely annotated settings.

③ **Team “VIM”** participated in Track 2, and they came from Nanjing University of Science and Technology. The team built their algorithmic framework based on AdaTAD [21], as illustrated in Figure 6. To enhance the modeling of channel-wise importance, they integrated an SE-Net block after the depthwise separable temporal convolutional layer within each backbone adapter. To further improve the model’s capacity for detecting short-duration actions, the team introduced a cross-attention fusion module into the feature pyramid. This module facilitated the injection of high-level semantic information into lower-level features. To strengthen the model’s classification performance, the team also selected representative samples from the MA-52 dataset and incorporated them into the training set during the early stages of model training. For implementation, the team adopted the VideoMAE-Large model as the backbone, which had been pre-trained on the Kinetics-700 dataset.

4 Challenge results and discussion

4.1 Track 1 Results Analysis

In the Micro-Action Recognition track, the top-3 performing teams and their results are shown in Table 3. The USTC-IAT-United team

achieved the highest overall performance with an F1_{mean} of 77.75, demonstrating balanced effectiveness across body and action levels. Although HIT obtained the best body-level scores and the highest action Top-5 accuracy, it ranked 2nd overall. The Wind of Innovation team delivered consistent performance across metrics, ranking 3rd. The close scores among the top teams underscore the competitiveness of the Micro-Action Recognition task. Notably, both HIT and Wind of Innovation adopted Large Language Models (LLMs) for model ensembling, showcasing a novel and promising direction for integrating LLMs into fine-grained action understanding.

4.2 Track 2 Results Analysis

In the Multi-label Micro-Action Detection track, the top-3 performing teams and their results are shown in Table 4. The USTC-IAT-United team achieved the best overall performance with an average mAP of 27.04, consistently outperforming others across all IoU thresholds. The NJU-MCG and VIM teams followed closely with average mAPs of 25.15 and 24.87, respectively, demonstrating strong capabilities in multi-label detection. All three teams adopted AdaTAD as the backbone. However, the overall performance remains relatively low, highlighting the inherent difficulty of accurately recognizing micro-action categories in the multi-label micro-action detection setting.

5 Conclusion and Future Directions

As a continuous event, the 2nd MAC Challenge has drawn increased participation from a broader range of teams and institutions, further advancing the field of micro-action analysis. In the future, we plan to extend the MAC Challenge in the following directions: 1). **Multimodal Micro-Action Analysis:** Combine RGB with skeleton, depth, and optical flow to capture richer motion cues and improve recognition robustness. 2). **Efficient Modeling:** Address the heavy reliance on large networks by promoting lightweight and real-time models suitable for deployment. 3). **Empowering Vision-Language Models (VLMs):** Explore the use of Large Vision-Language Models (LVLMs) to understand and reason about micro-actions through multimodal inputs and in-context learning.

¹The page for Track 1: <https://codalab.lisn.upsaclay.fr/competitions/22399>

²The page for Track 2: <https://codalab.lisn.upsaclay.fr/competitions/22455>

Acknowledgments

The author gratefully acknowledges the support of K. C. Wong Education Foundation. The author sincerely thanks the organizers of ACM MM 2025 for their support in hosting our challenge. This work is supported by National Key R&D Program of China (2024YFB3311600), Natural Science Foundation of China (72188101, 62272144, and 62020106007), the Anhui Provincial Natural Science Foundation (2408085J040), the Fundamental Research Funds for the Central Universities (JZ2024HGTG0309, JZ2024AHST0337, and JZ2023YQTD0072), and the New Cornerstone Science Foundation through the XPLOER PRIZE.

References

- [1] Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhulius, et al. 2025. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985* (2025).
- [2] Michal Balazsia, Philipp Müller, Ákos Levente Tanczos, August von Liechtenstein, and Francois Bremond. 2022. Bodily behaviors in social interaction: Novel annotations and state-of-the-art evaluation. In *Proceedings of the 30th ACM International Conference on Multimedia*. 70–79.
- [3] Joao Carreira and Andrew Zisserman. 2017. Quo vadis, action recognition? a new model and the kinetics dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 6299–6308.
- [4] Guoliang Chen, Fei Wang, Kun Li, Zhiliang Wu, Hehe Fan, Yi Yang, Meng Wang, and Dan Guo. 2024. Prototype learning for micro-gesture classification. *arXiv preprint arXiv:2408.03097* (2024).
- [5] Haoyu Chen, Björn W Schuller, Ehsan Adeli, and Guoying Zhao. 2024. The 2nd Challenge on Micro-gesture Analysis for Hidden Emotion Understanding (MiGA) 2024: Dataset and Results. In *MiGA 2024: Proceedings of IJCAI 2024 Workshop & Challenge on Micro-gesture Analysis for Hidden Emotion Understanding (MiGA 2024) co-located with 33rd International Joint Conference on Artificial Intelligence (IJCAI 2024)*. RWTH Aachen.
- [6] Haoyu Chen, Henglin Shi, Xin Liu, Xiaobai Li, and Guoying Zhao. 2023. SMG: A micro-gesture dataset towards spontaneous body gestures for emotional stress state analysis. *International Journal of Computer Vision* 131, 6 (2023), 1346–1366.
- [7] Fan Gong, Jialiang Chen, Jiajun Zhu, Qijian Bao, Fei Gao, Renshu Gu, and Gang Xu. 2024. Micro-action recognition via hierarchical fusion and inference. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 11327–11332.
- [8] Jihao Gu, Kun Li, Fei Wang, Yanyan Wei, Zhiliang Wu, Hehe Fan, and Meng Wang. 2025. Motion matters: Motion-guided modulation network for skeleton-based micro-action recognition. *arXiv preprint arXiv:2507.21977* (2025).
- [9] Jihao Gu, Fei Wang, Kun Li, Zhiliang Wu, and Dan Guo. 2025. MM-Gesture: Towards Precise Micro-Gesture Recognition through Multimodal Fusion. *arXiv preprint arXiv:2507.08344* (2025).
- [10] Dan Guo, Kun Li, Bin Hu, Yan Zhang, and Meng Wang. 2024. Benchmarking Micro-action Recognition: Dataset, Methods, and Applications. *IEEE Transactions on Circuits and Systems for Video Technology* 34, 7 (2024), 6238–6252.
- [11] Dan Guo, Xiaobai Li, Kun Li, Haoyu Chen, Jingjing Hu, Guoying Zhao, Yi Yang, and Meng Wang. 2024. MAC 2024: Micro-Action Analysis Grand Challenge. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 11304–11305.
- [12] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276* (2024).
- [13] Kun Li, Dan Guo, Guoliang Chen, Chunxiao Fan, Jingyuan Xu, Zhiliang Wu, Hehe Fan, and Meng Wang. 2025. Prototypical calibrating ambiguous samples for micro-action recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 4815–4823.
- [14] Kun Li, Dan Guo, Guoliang Chen, Xinge Peng, and Meng Wang. 2023. Joint skeletal and semantic embedding loss for micro-gesture classification. *arXiv preprint arXiv:2307.10624* (2023).
- [15] Kun Li, Pengyu Liu, Dan Guo, Fei Wang, Zhiliang Wu, Hehe Fan, and Meng Wang. 2025. MMAD: Multi-label Micro-Action Detection in Videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- [16] Kun Li, Xinge Peng, Dan Guo, Xun Yang, and Meng Wang. 2025. Repetitive Action Counting with Hybrid Temporal Relation Modeling. *IEEE Transactions on Multimedia* 27 (2025), 3844–3855.
- [17] Qiankun Li, Qiupu Chen, Huabao Chen, Feng He, Depeng Li, and Zhigang Zeng. 2025. Progressive Large-Scale Modeling via Temporal-Spatial Focus Connector for Micro-Action Recognition. In *Proceedings of the 33rd ACM International Conference on Multimedia*.
- [18] Qiankun Li, Xiaolong Huang, Huabao Chen, Feng He, Qiupu Chen, and Zengfu Wang. 2024. Advancing micro-action recognition with multi-auxiliary heads and hybrid loss optimization. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 11313–11319.
- [19] Qiankun Li, Xiaolong Huang, Zhifan Wan, Lanqing Hu, Shuzhe Wu, Jie Zhang, Shiguang Shan, and Zengfu Wang. 2023. Data-efficient masked video modeling for self-supervised action recognition. In *Proceedings of the 31st ACM International Conference on Multimedia*. 2723–2733.
- [20] Pengyu Liu, Fei Wang, Kun Li, Guoliang Chen, Yanyan Wei, Shengeng Tang, Zhiliang Wu, and Dan Guo. 2024. Micro-gesture Online Recognition using Learnable Query Points. *arXiv preprint arXiv:2407.04490* (2024).
- [21] Shuming Liu, Chen-Lin Zhang, Chen Zhao, and Bernard Ghanem. 2024. End-to-end temporal action detection with 1b parameters across 1000 frames. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18591–18601.
- [22] Xin Liu, Henglin Shi, Haoyu Chen, Zitong Yu, Xiaobai Li, and Guoying Zhao. 2021. imigue: An identity-free video dataset for micro-gesture understanding and emotion analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10631–10642.
- [23] Adrien Pavao, Isabelle Guyon, Anne-Catherine Letournel, Dinh-Tuan Tran, Xavier Baro, Hugo Jair Escalante, Sergio Escalera, Tyler Thomas, and Zhen Xu. 2023. CodaLab Competitions: An Open Source Platform to Organize Scientific Challenges. *Journal of Machine Learning Research* 24, 198 (2023), 1–6. <http://jmlr.org/papers/v24/21-1436.html>
- [24] Wei Qian, Kun Li, Dan Guo, Bin Hu, and Meng Wang. 2024. Cluster-phys: Facial clues clustering towards efficient remote physiological measurement. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 330–339.
- [25] Yunjie Tian, Qixiang Ye, and David Doermann. 2025. Yolov12: Attention-centric real-time object detectors. *arXiv preprint arXiv:2502.12524* (2025).
- [26] Chuang Wang, Weidong Chen, Xu Cui, Yiming Zhao, Zhaobo Qi, Pengqi Huang, Xinyan Liu, and Weigang Zhang. 2025. Combatting Data Imbalance and Noise in Micro-Action Recognition. In *Proceedings of the 33rd ACM International Conference on Multimedia*.
- [27] Chen Wang, Xun Mei, and Feng Zhang. 2024. Instance-aware fine-grained micro-action recognition. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 11320–11326.
- [28] Fei Wang, Dan Guo, Kun Li, and Meng Wang. 2024. Eulermormer: Robust eulerian motion magnification via dynamic filtering within transformer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 5345–5353.
- [29] Fei Wang, Dan Guo, Kun Li, Zhun Zhong, and Meng Wang. 2024. Frequency decoupling for motion magnification via multi-level isomorphic architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18984–18994.
- [30] Fei Wang, Kun Li, Yiqi Nie, Zhangling Duan, Peng Zou, Zhiliang Wu, Yuwei Wang, and Yanyan Wei. 2025. Exploiting ensemble learning for cross-view isolated sign language recognition. In *Companion Proceedings of the ACM on Web Conference 2025*. 2453–2457.
- [31] Limin Wang, Bingkun Huang, Zhiyu Zhao, Zhan Tong, Yinan He, Yi Wang, Yali Wang, and Yu Qiao. 2023. Videomae v2: Scaling video masked autoencoders with dual masking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 14549–14560.
- [32] Limin Wang, Yuanjun Xiong, Zhe Wang, Yu Qiao, Dahua Lin, Xiaoou Tang, and Luc Van Gool. 2016. Temporal segment networks: Towards good practices for deep action recognition. In *Proceedings of European Conference on Computer Vision*. 20–36.
- [33] Zhiliang Wu, Kerui Chen, Kun Li, Hehe Fan, and Yi Yang. 2025. BVINet: Unlocking blind video inpainting with zero annotations. *arXiv preprint arXiv:2502.01181* (2025).
- [34] Zhichao Xia, Yichi Zhang, Yanjun Chi, Lingsi Zhu, Mohan Jing, and Jun Yu. 2025. Hierarchical Multi-Feature Extraction and Aggregation for Micro-Action Recognition. In *Proceedings of the 33rd ACM International Conference on Multimedia*.
- [35] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388* (2025).
- [36] Jun Yu, Mohan Jing, Guopeng Zhao, Keda Lu, Yifan Wang, Feng Zhao, Jiaqing Sun, Qingsong Liu, and Jiaen Liang. 2024. End-to-end Spatio-Temporal Information Aggregation For Micro-Action Detection. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 11306–11312.
- [37] Guoying Zhao, Xiaobai Li, Yante Li, and Matti Pietikäinen. 2023. Facial micro-expressions: An overview. *Proc. IEEE* 111, 10 (2023), 1215–1235.
- [38] Jiaqi Zhao, Fei Wang, Kun Li, Yanyan Wei, Shengeng Tang, Shu Zhao, and Xiao Sun. 2025. Temporal-Frequency State Space Duality: An Efficient Paradigm for Speech Emotion Recognition. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing*. 1–5.
- [39] Yue Zhu, Kun Li, and Zongxin Yang. 2025. Exploiting EfficientSAM and Temporal Coherence for Audio-Visual Segmentation. *IEEE Transactions on Multimedia* 27 (2025), 2999–3008.